

1.3 SET OPERATIONS

There is an analogy between set theory and real number theory. In real number theory we study four main operations, viz. addition, subtraction, multiplication and division. Similar algebraic operations can be defined for sets as well to facilitate the study of the properties of certain combinations of sets. While defining these operations we only consider elements belonging to an already chosen universal set S .

Complement of a Set

If A is any subset of the universal set S , the set consisting of the elements in S that do not belong to A is called the *complement* of A . The complement of A is denoted by A' . Thus, $A' = \{x \mid x \notin A\}$.

Example 1.3.1

If $S = \{1, 2, 3, 4, 5\}$ and $A = \{1, 3, 5\}$, then $A' = \{2, 4\}$.

Example 1.3.2

If S is the set of all bolts produced by a machine and A denotes the set of all bolts of accepted quality, A' denotes the set of all defective bolts.

The complement S' of the universal set is clearly ϕ . We need to know the universal set S to define the complement of a subset A . The definition of the complement of a set is symmetric in the sense that the "complement of the complement of a set is the original set itself". In symbols, this is expressed as $(A')' = A$.

Intersection of Sets

If A and B are any two subsets of S , we can define another set as the set of all those elements that belong to *both* A and B . This new set is called the *intersection* of A and B and is denoted by $A \cap B$. In symbols

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$$

Example 1.3.3

If A is the set of all subscribers to "Illustrated Monthly" and B is the set of all subscribers to "Suvidha", the set $A \cap B$ is the set of all persons who subscribe to "Illustrated Monthly" and "Suvidha".

Example 1.3.4

Let A be the set of all persons in a city whose annual income is above

Rs 10,000 and B be the set of all persons in the same city whose annual income is between Rs 7,000 and Rs 8,000. $A \cap B$ in this case is the null set, since no person can be in both the sets A and B .

If A and B are two sets such that $A \cap B = \phi$, i.e. there are no elements common to both A and B , we say that A and B are *disjoint* (mutually exclusive) sets. For example, any set A and its complement A' are disjoint sets.

Union of Sets

If A and B are any two sets, the set of all elements that belong to either A or B or both A and B is called the *union* of A and B and is denoted by $A \cup B$.

Example 1.3.5

Let S be the set of all married couples in a locality.

Let

A = Set of all couples in which husbands are older than 45 years in age

B = Set of all couples in which husbands are older than their wives

C = Set of couples with wives older than 45 years in age.

$A \cup C$ is the set of all couples with at least one of them older than 45 years in age. $A \cap B$ is the set of all couples in which husbands are older than wives and husbands are older than 45 years in age.

Example 1.3.6

If A and B are defined as in Example 1.3.3, $A \cup B$ represents all those who subscribe to at least one of the two magazines.

Note that $A \cap B \subset A \cup B$, since the elements that belong to both A and B also belong to $A \cup B$. Further, $A \cup A' = S$, for any set A .

Difference of Two Sets

If A and B are any two sets, the set $A - B$ is the set of all elements that belong to A but not to B . In symbols,

$$A - B = \{x \mid x \in A; x \notin B\}$$

$A - B$ can also be considered as the complement of B relative to A . If A is the set of all income tax payers and B is the set of all wealth tax payers $A - B$ represents income tax payers who do not pay wealth tax.

The abstract concepts relating to sets can be illustrated by pictorial diagrams. These diagrams are often called *Venn diagrams*. We use a rectangle to represent the universal set. A circle drawn inside the rectangle will represent a set. We make circles overlap one another if their intersection is not empty. The points inside the circle represent the elements of the corresponding set. The Venn diagrams illustrating the concept of complement, union, intersection, and difference of two sets are shown in Fig. 1.3.1.

The operations of unions and intersections may be extended to any number of sets. We define the intersection of a collection of sets as the set which contains those elements which belong to all the sets. The union of a

collection of sets is the set of those elements which belong to at least one of the sets in the collection. Thus, if A, B, C are any three sets,

$$A \cap B \cap C = \{x \mid x \in A, x \in B, x \in C\}$$

$$A \cup B \cup C = \{x \mid x \text{ belongs to at least one of } A \text{ or } B \text{ or } C\}$$

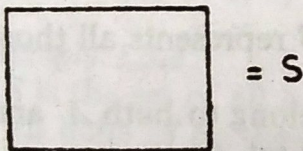
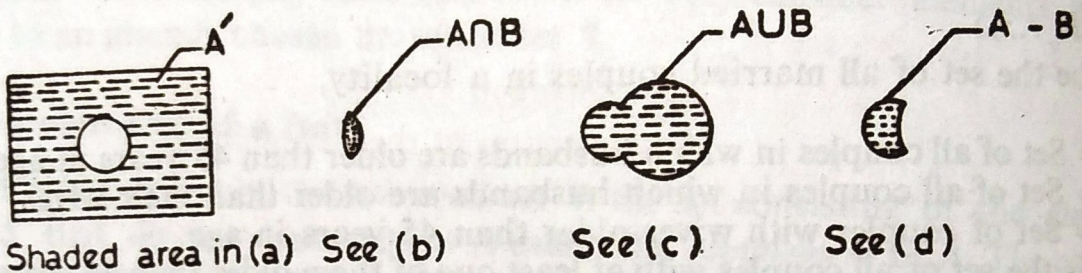
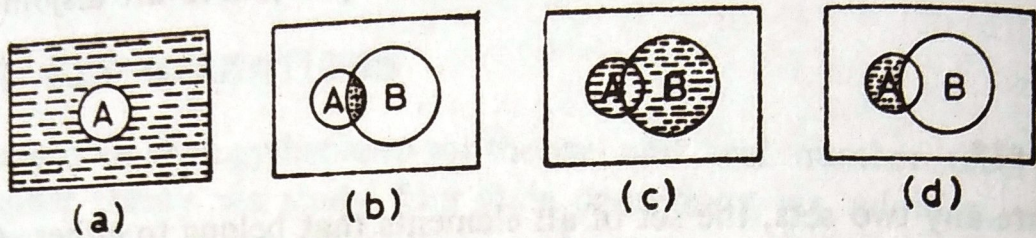


Fig. 1.3.1

1.4 FUNDAMENTAL LAWS OF SET OPERATION

Just as there are laws for the operation of numbers in the real number system, there are some basic laws of set operation also. We list below some of these laws. Most of these laws follow directly from definition or they can be verified by drawing the Venn diagrams. All sets defined below are the subsets of the universal set S .

(i) Identity law

$$A \cup \phi = A$$

$$A \cap \phi = \phi$$

$$A \cup S = S$$

$$A \cap S = A$$

(ii) Complement law

$$A \cup A' = S$$

$$A \cap A' = \phi$$

$$(A')' = A$$

(iii) Idempotent law

$$A \cup A = A$$

$$A \cap A = A$$

(iv) Associative law

$$(A \cup B) \cup C = A \cup (B \cup C) \quad (A \cap B) \cap C = A \cap (B \cap C)$$

(v) Commutative law

$$A \cup B = B \cup A \quad A \cap B = B \cap A$$

(vi) Distributive law

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

(vii) Law of inclusion

$$A \subset (A \cup B) \quad B \subset (A \cup B)$$

$$(A \cap B) \subset A \quad (A \cap B) \subset B$$

(viii) Law of difference

$$A - \phi = A$$

$$A - A = \phi$$

(x) De Morgan's laws

$(A \cup B)' = A' \cap B'$, i.e. complement of union of two sets is the intersection of their complements.

$(A \cap B)' = A' \cup B'$, i.e. complement of intersection of two sets is the union of their complements.

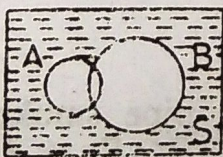
More generally, if $A_1, A_2, \dots, A_n$ are any sets,

$$(A_1 \cup A_2 \cup \dots \cup A_n)' = A_1' \cap A_2' \cap A_3' \dots \cap A_n'$$

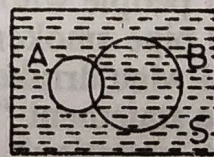
$$(A_1 \cap A_2 \cap \dots \cap A_n)' = A_1' \cup A_2' \cup \dots \cup A_n'$$

Only the first of the two De Morgan's laws is verified here through Venn diagram (see Fig. 1.4.1). The reader is urged to verify the other law.

Verification of $(A \cup B)' = A' \cap B'$

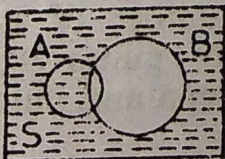


(a)

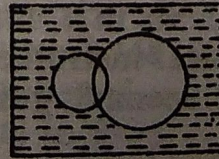


(b)

Non-shaded portion: $A \cup B$ Non-shaded region: A
Shaded portion: $(A \cup B)'$ Shaded region: A'



(c)



(d)

Non-shaded region: B Shaded region: $A \cap B'$
Shaded region: B'

(a) and (d) are the same

Fig. 1.4.1

From the Figs 1.4.1 (a) and (d) it is clear that $A' \cap B'$ is precisely equal to the shaded region of Fig. (a) which is $(A \cup B)'$.

We can prove rigorously the same fact using the definition of set equality given in Sec. 1.2.

Proof of $(A \cup B)' = A' \cap B'$

This is done by showing that $(A \cup B)' \subset A' \cap B'$ and simultaneously $A' \cap B' \subset (A \cup B)'$. Let x be any element in $(A \cup B)'$. Since x is a member of the complement of $A \cup B$, x cannot belong to A and also to B . Hence, x belongs to A' as well as to B' . Thus, $x \in A' \cap B'$. Thus, every member of $(A \cup B)'$ is also a member of $A' \cap B'$. This means $(A \cup B)' \subset A' \cap B'$. Conversely, take any element $y \in A' \cap B'$. Since y belongs to both A' and B' , it does not belong either to A or to B . In other words, y belongs to the complement of $A \cup B$, i.e. $y \in (A \cup B)'$. Thus, every member of $A' \cap B'$ is a member of $(A \cup B)'$. This means $(A' \cap B') \subset (A \cup B)'$. This, in conjunction with the earlier established result $(A \cup B)' \subset (A' \cap B')$ proves the desired result $(A \cup B)' = A' \cap B'$ by the definition of set equality.

We shall give a few examples on set constructions and the use of Venn diagrams.